Experience Admission for White-Box Agentic RL: A Position on Dataflow between Exploration and the Base Policy

Anonymous Author(s)
Affiliation
Address
email

Abstract

In white-box instrumented-worker multi-worker RL, a layer is often missing between optimizers and collaboration surfaces: after trajectories are produced, who may materialize, at what depth, and when may θ update. Black-box APIs are out of strong-form claim.

We call this layer experience admission. Novelty is P1–P4 (a proposed admission policy) plus preregistered failure conditions—not a new score or proven discovery. Hypothesis-level claim: the bottleneck is often how experience enters the base update path in a layered, isolated, low-frequency way.

Layered test: fix $\mathcal{A}$; vary only the data plane (A/B/P). If failures 4–5 do not fire, materialization cost falls vs. A, and the task is not in persistent regression, meet the scale-up gate only (allows larger runs; $\neq$ validation; does not alone entail "must ship as a first-class module"). After scale-up still passes, then recommend a first-class pipeline module; else withdraw that configuration's strong form.

1 Motivation (scoped)

As agent environments scale—more workers, longer horizons, denser trajectory streams—three layers are often mixed: optimizer (when/how to update; e.g., group-relative methods [1]; single-rollout async optimization [2]), collaboration (shared context at interaction time), and admission (who may materialize, how deeply, when θ may be touched). Stability $\neq$ admission: the former asks whether updates diverge on a visible set; the latter asks which trajectories may enter that set and whether/how they are materialized first. IMPALA [3] can still treat "delivered $\Rightarrow$ learnable." Shielding [4] constrains actions; CQL [5] constrains offline values—not online materialization/update budgets. Scaling the environment without an admission layer can amplify unbudgeted exposure of noise and hazard into the base policy.

Minimal counterexample. With ordinary learnable τ^{+} and high-TD-error hazard $\tau^{\ddagger}$ in one buffer, PER [6] allows $\tau^{\ddagger}$ into the learner above uniform rate when high-priority (power-law sampling: high quantile does not guarantee a draw, but the mechanism does not exclude hazard uplift). Action shields do not remove it from the train-visible set; delete filters discard analysis value. Gap triad: P1 budget semantics absent (pool-default learnable; no default-off materialization), plus P2·P4 / P3 violations. If hazard has low TD-error, PER may ignore it—that is "unnoticed," not policy-level quarantine-with-retention (P2). Sampling weights $\neq$ admission policy.

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

Table 1: Proposed constraints P1–P4 vs. typical PER / delete-style filters.

	Meaning	vs PER / delete-filter
P1	Full materialization off by default (explicit quota)	Usually absent
P2	Quarantine $\neq$ delete	Usually absent
P3	Full materialization $\neq$ admission	Sampled $\Rightarrow$ learned
P4	$D_{\text{adm}} \subsetneq D_{\text{all}}$ when hot nonempty	Reweight-all / action-only

Scope. Strong form assumes workers expose hidden representations, local action distributions, or uncertainty signals. Fingerprints and thresholds are knobs, not results. Black-box APIs degrade to post-hoc text filtering and must not be conflated with the strong form.

2 Proposed admission policy (not "irreducible discovery")

Terms. Materialization = persist a trajectory from a streaming state into an inspectable object; deep read = costly analysis on (usually) materialized trajectories. They often share an admission-cost budget but are logically separable. For brevity we say materialization cost in budget/metrics. D_{read} is eligible for materialize/deep-read, not admit-to-update.

Use the name only if all four are enforced; else revert terms. P1–P4 are a proposed admission policy, not an established scientific finding.

Logical interdependence (design-level, not ablation). Drop P1 $\rightarrow$ materialization budget vanishes; warm/hot become post-hoc labels $\approx$ unbudgeted PER + tags. Drop P2 $\rightarrow$ delete-or-learn (no analysis track). Drop P3 $\rightarrow$ $D_{\text{read}} \equiv D_{\text{adm}} \approx$ PER/passive pool. Drop P4 $\rightarrow$ P1–P3 may remain, but when hot is nonempty $D_{\text{adm}} \subsetneq D_{\text{all}}$ is no longer forced: a hybrid cache still distinct from PER, yet incomplete as a visible-set policy. Together they block four distinct collapses; joint proposal is warranted. This does not replace causal ablations (§5).

LLM curation (filtering / RLAIF / RAFT [7–9]) optimizes pre-batch quality; kept $\Rightarrow$ update θ. High reward $\neq$ low risk. Multi-phase exploration (e.g., First Return, Then Explore [10]) targets exploration schedules and coverage—orthogonal to materialization budgets, quarantine tracks, or enforced $D_{\text{adm}} \subsetneq D_{\text{all}}$.

Honest empirical gap: no causal ablation of which constraint drives gains vs. PER. H3 (measurable quarantine utility) is exploratory for the scale-up gate; P2 execution integrity (nonempty quarantine must be read/used) is enforced by failure 4(b).

H$\leftrightarrow$P: H1$\rightarrow$P1 (primary); H2$\rightarrow$P2+P4 (primary); H3$\rightarrow$P2 value magnitude (exploratory); H4 (contact frequency / visible-set size, not SGD step size)$\rightarrow$P3+P4 (primary).

3 Interface, predicate, contract

Not a full formal system—an interface plus minimal predicates. Dataflow (pool $\mathcal{P}$, not batch B_t):

$$\tau \xrightarrow{\phi} x \xrightarrow{r} \mathcal{P} \xrightarrow{\text{adm}} D_{\text{adm}} \xrightarrow{\mathcal{A}} \theta.$$

$$\tau \in D_{\text{adm}} \iff \text{Validated}(\tau) \wedge \neg\text{HotHazard}(\tau) \wedge \text{InBudget}(\tau),$$

with HotHazard $\iff$ (r=hot $\wedge$ ℓ=$\texttt{hazard}$). Operational floor for Validated: promotion must pass a check independent of the routing score that proposed promotion (held-out probe, second scorer, or human/audit gate—fixed in the experimental plan). Hazard labels must not be dictated solely by the same routing score. Interaction: every optimizer batch $B_t \subseteq D_{\text{adm}}$. $B_t \not\subseteq D_{\text{adm}}$ is a contract breach; if the batch contains hot-hazard, failure 4 also fires.

Contract: (1) no hot-hazard in update batches; (2) materialization $\neq$ write θ; (3) promotion validated independently of routing score; (4) archive closed by default; warm budgeted; (5) optional curator cannot enlarge allowed sets. Strong form forbids admit-without-materialization as default.

4 Preregistered falsification

Arms: A passive (default materializable / replay); B admission (P1–P4+§3); P optional PER (weights $\neq$ admission).

Metrics: materialization ratio; wall-clock/FLOPs; return; hazard penetration; materialization gain = probe return lift or loss drop after materialization (pick one; fixed in plan)—subset mean/median or preregistered smoothing allowed; Spearman ρ vs. (aggregated) gain for failure-5 labels only.

Failures ($K \geq 3$ segments; any one rejects the setting):

1. Warm-mass collapse—warm mass (routing count) $\geq 90\%$ (or archive+hot $< 10\%$) and materialization improvement vs. A $< 5\%$ (conjunction; high mass with clear cost drop does not fire).
2. No budget gain—materialization improvement $< 10\%$ and wall-clock/FLOPs improvement $< 5\%$ (default AND). Grey zone: mat. gain in $[5\%, 10\%)$ with wall-clock $\geq 5\%$ fires neither 1–2 nor meets the scale-up gate—no scale-up claim; alternative joint criteria require preregistration.
3. Persistent regression vs. A (preregistered α/MID).
4. Hard (P2 integrity): (a) any hot-hazard enters D_{adm} and updates θ; or (b) quarantine/hot nonempty in-window but never read/deep-read and unused for any preregistered peripheral analysis ("quarantine-to-forget"). (b) N/A if hot empty.
5. Hard proxy–gain test (three clauses; current proxy family only; any clause firing fails—run all three):
   - Correlation: one-sided Spearman of $\rho > 0$ at preregistered α (e.g. 0.05); nonsignificant $\rightarrow$ fires.
   - Random baseline: archive F1/accuracy within preregistered δ of a matched-marginal i.i.d. random archive labeling $\rightarrow$ fires.
   - Family boundary: "new family" = architectural score change (e.g. entropy$\rightarrow$prediction error); retuning thresholds/windows inside the same function = same instance (must retest). After fail: at most weak post-hoc filtering; no strong-form mix; new family requires fresh preregistration.

Scale-up gate ($\neq$ validation): materialization $\downarrow\geq 10\%$ vs. A; failure 3 unmet; failures 4–5 unmet (incl. 4(b)). Allows larger experiments only—does not alone entail a first-class module. After scale-up still passes: then recommend first-class pipeline module. H3 value magnitude remains exploratory.

5 Limitations

No theory; no causal minimality claim; no black-box strong form; no LoRA/GRPO main results; no appendix-as-proof. Condition 5 is intentionally strict: under small N or high label noise, nonsignificance is likely and should be read as "this proxy family is unsupported," not "the admission question is void." Materialization-gain labels are measured on trajectories selected for materialization, so selection bias is intrinsic; the random-baseline clause partially disciplines it but does not remove it. Scope boundary: long-horizon, costly materialization/deep-read, heterogeneous workers; short trajectories / single worker / negligible cost may favor passive pool or PER—even after a pass, designers may decline to ship. Future: ablations; open-weight multi-seed; learnable routers; H3 value metrics beyond 4(b); separate paper for black-box weak form [11–13].

6 Conclusion

Deliverables: name the (white-box) layer; propose P1–P4 with logical interdependence; give a falsifiable protocol. Scale-up gate $\neq$ validation and $\neq$ must-ship; after scale-up still passes, then recommend a first-class pipeline module. Whether the hypothesis holds at scale remains open—outside any "already proven" claim.

References

[1] Z. Shao, P. Wang, Q. Zhu, et al. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models. arXiv preprint arXiv:2402.03300, 2024.

[2] Z. Hou, Y. Li, J. Tang, and Y. Dong. Single-rollout asynchronous optimization for agentic reinforcement learning. arXiv preprint arXiv:2607.07508, 2026.

[3] L. Espeholt et al. IMPALA: Scalable distributed deep-RL with importance weighted actor-learner architectures. In International Conference on Machine Learning (ICML), 2018.

[4] M. Alshiekh et al. Safe reinforcement learning via shielding. In Proceedings of the AAAI Conference on Artificial Intelligence, 2018.

[5] A. Kumar, A. Zhou, G. Tucker, and S. Levine. Conservative Q-learning for offline reinforcement learning. In Advances in Neural Information Processing Systems (NeurIPS), 2020.

[6] T. Schaul, J. Quan, I. Antonoglou, and D. Silver. Prioritized experience replay. In International Conference on Learning Representations (ICLR), 2016.

[7] Y. Hu et al. Towards comprehensive preference data collection for reward modeling. arXiv preprint arXiv:2406.16486, 2024.

[8] H. Lee et al. RLAIF vs. RLHF. arXiv preprint arXiv:2309.00267, 2023.

[9] H. Dong et al. RAFT: Reward ranked finetuning. Transactions on Machine Learning Research, 2023.

[10] A. Ecoffet et al. First return, then explore. Nature, 590:580–586, 2021.

[11] T. Shen et al. A comprehensive survey of LLM alignment techniques. arXiv preprint arXiv:2407.16216, 2024.

[12] D. Pathak, P. Agrawal, A. Efros, and T. Darrell. Curiosity-driven exploration by self-supervised prediction. In International Conference on Machine Learning (ICML), 2017.

[13] Y. Burda, H. Edwards, A. Storkey, and A. Klimov. Exploration by random network distillation. In International Conference on Learning Representations (ICLR), 2019.